

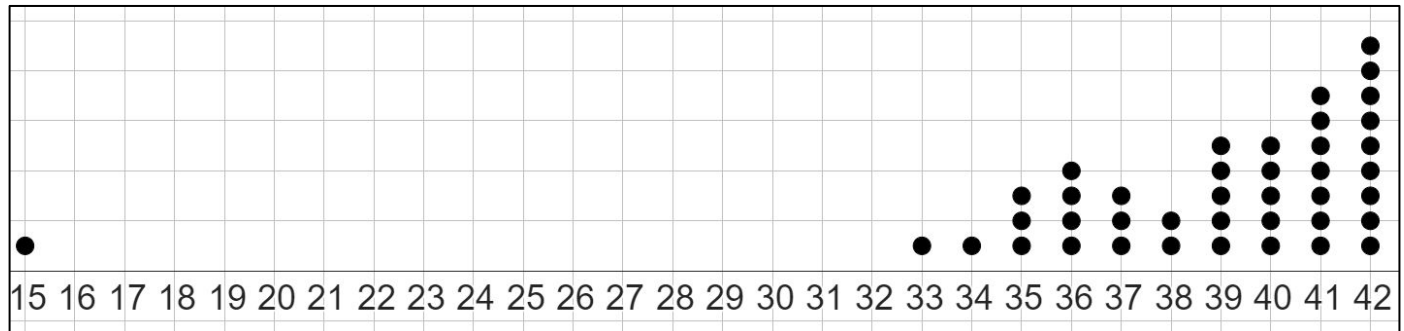
Name: _____



Evaluating the range

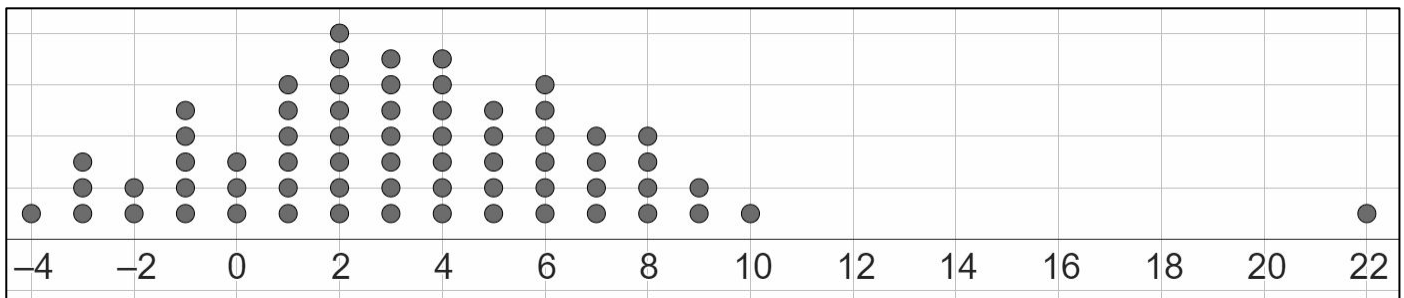
Task A: Outliers and their effect on the range

1)

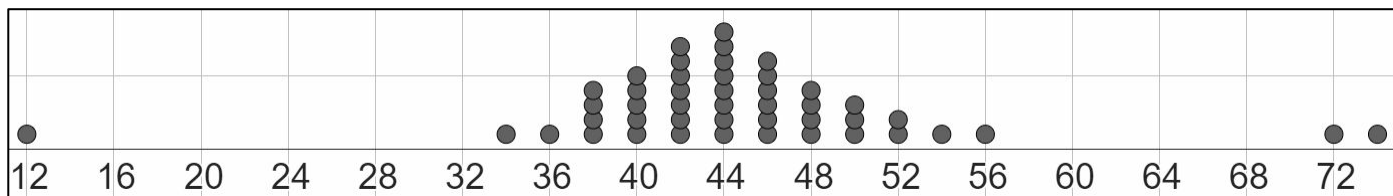


- Circle the **outlier** in this dataset.
- What is the original range of this dataset?
- If a data point for the number 39 was removed, by how much would the range change? Explain your answer.
- Remove the **outlier** from the dataset. Calculate the new range.

2)



- Circle the **outlier** in this dataset.
- What is the original range of this dataset?
- If a data point for the number -4 was removed, by how much would the range change? Explain your answer.
- Remove the **outlier** from the dataset. Calculate the new range.



3) All data points in this dataset are even numbers.

a) What are the values of the three **outliers**?

b) By how much would the range decrease by if all the **outliers** were removed?

1.5	14.5	16.9	21	24.2	28.3	31.9	33.8	37.7	38.8	39.9	42.5	45.2	46.8	82.5
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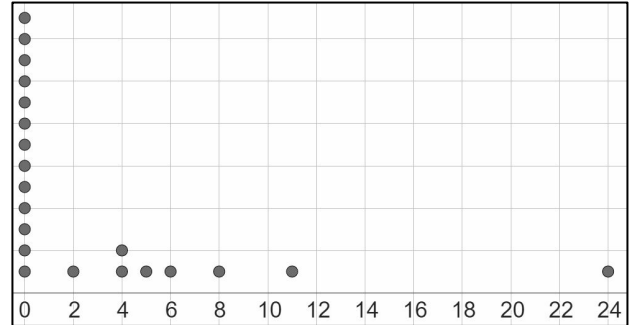
4) Complete each sentence with a number from the dataset:

- a) the original range is
- b) removing the data point 37.7 will change the new range to
- c) if the smaller **outlier** was removed from the original dataset, the new range would be
- d) if the larger **outlier** was removed from the original dataset, the new range would be
- e) the **outlier** that would be more impactful to remove is



Task B: Evaluating if removing outliers is sensible

1) This dot plot shows the number of days per month that had airfrost (due to cold weather) in Leuchars (Scotland) over a 21 month period.

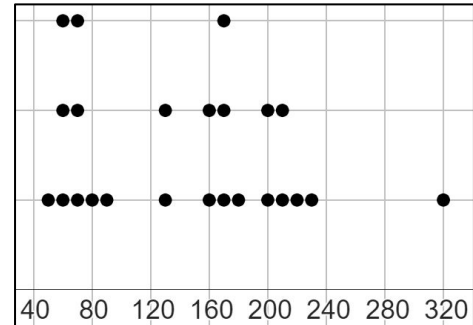


a) What is the value of the **outlier**?

b) Should the **outlier** be: removed, kept, or investigated further?

Write a sentence to explain the option you have chosen.

2) This dot plot shows the number of “hours of sunshine” per month for two years, rounded to the nearest 10 hours.



a) What is the value of the **outlier**?

b) All data points have been correctly measured, with no error.

Should the **outlier** be: removed, kept, or investigated further?

Write a sentence to explain the option you have chosen.

3) This dataset shows the height of 24 chimpanzees, in centimetres.

1.4	95	103	108	108	111	114	117
118	118	124	125	129	132	135	138
130	138	150	153	161	166	172	198

a) What is the current range of this dataset?

b) What is the value of the **outlier** or **outliers** in this dataset?

c) Should the **outlier** or **outliers** be: removed, kept, or investigated further?

Write a sentence to explain the option you have chosen, and justify it by calculating any new range that might happen from you removing any **outliers**.



4) This dataset accurately shows the total monthly rainfall (in millimetres) each month for two years, between 2007 and 2008.

Jun wants to represent this dataset on a dot plot, and wants to see if removing any **outliers** has an impact on the range of the data he has to plot.

Calculate two ranges, with one keeping the **outliers**, and one removing an **outlier**, and suggest if Jun should represent all data points on a dot plot or not.

285.6	60
154	58.3
146.9	57.5
113	56.7
108.7	52.8
98.8	48.7
93.2	44.5
92.4	43.5
83.8	28.1
82.6	27.1
79.2	23.3
74.9	5.8

